A categorical account of the Metropolis-Hastings algorithm

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March 4, 2026

Paper

A categorical account of the Metropolis–Hastings algorithm

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Background and motivation

Key question

Can the Metropolis-Hastings algorithm be understood using categorical probability?

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Can the Metropolis-Hastings algorithm be understood using categorical probability?

Two key motivations:

1. From a categorical perspective, provides a test case for recent developments (e.g. Markov categories)
2. From a statistics perspective, we would like a more conceptual language than measure theory

Basic concepts from Markov chain Monte Carlo

Key idea

Often in statistics (etc) we care about some integral

$$\int f(x)\,\mu(dx)$$

where μ is a probability measure on some space $\mathcal{X}$, and $f : \mathcal{X} \to \mathbb{R}$ is some function

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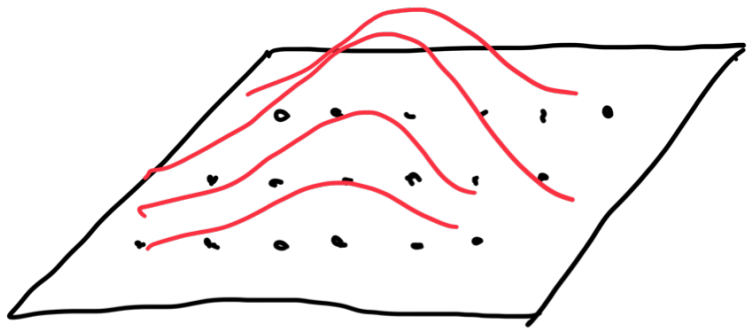
Usually this is intractable, so we must estimate it numerically

Naive approach

One approach is approximate via a Riemann sum, i.e.

$$\int f(x)\,\mu(dx) = \int f(x)\,\mu(x)\,dx \approx \frac{1}{N}\sum_{i=1}^{N} f(x_i)\,\mu(x_i)$$

for some chosen points $x_1, \ldots, x_N \in \mathcal{X}$



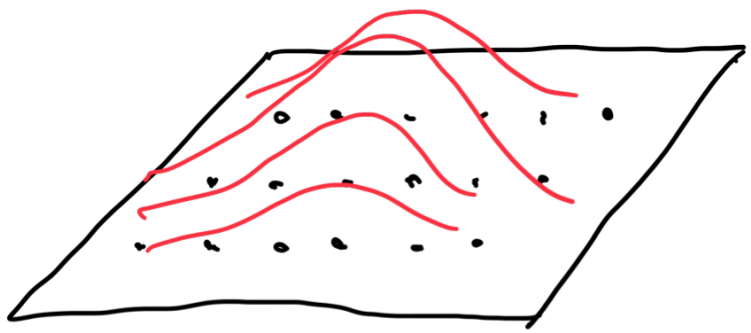
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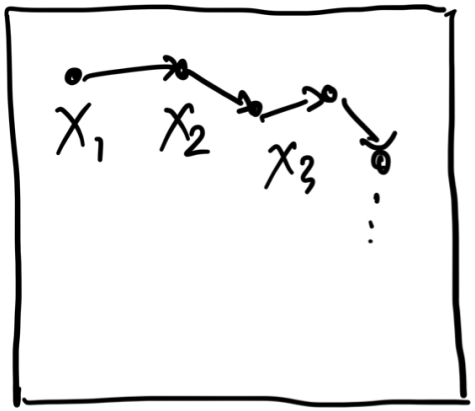


However, often very inefficient (and sensitive to choice of points)

Note: here we are assuming μ has a density $\mu(x)$, i.e. $\mu(dx) = \mu(x)\,dx$.

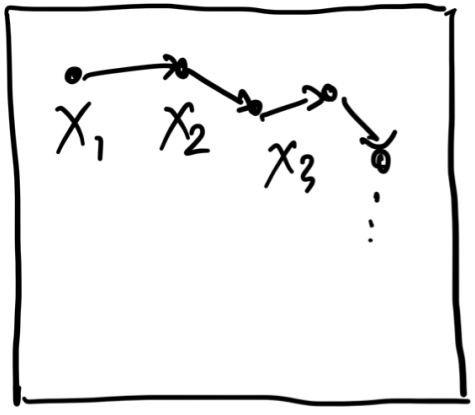
Markov chain Monte Carlo: idea

The idea of Markov chain Monte Carlo (MCMC) is to replace the grid by a Markov chain $(X_n)_{n\geq 1}$



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Formally, choose some initial point x_0, and generate

$$X_1 \sim P(x_0, \mathrm{d}x), \quad X_2 \sim P(X_1, \mathrm{d}x), \quad X_3 \sim P(X_2, \mathrm{d}x), \quad \ldots$$

where $K : \mathcal{X} \to \mathcal{X}$ is some chosen Markov kernel

Markov Chain Monte Carlo

Theorem

If P is chosen correctly, then under certain conditions $(X_n)_{n\geq 1}$ satisfies

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(X_n) \overset{\text{a.s.}}{=} \int f(x)\, \mu(dx).$$

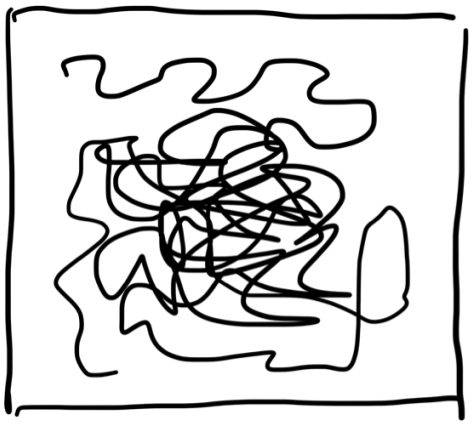
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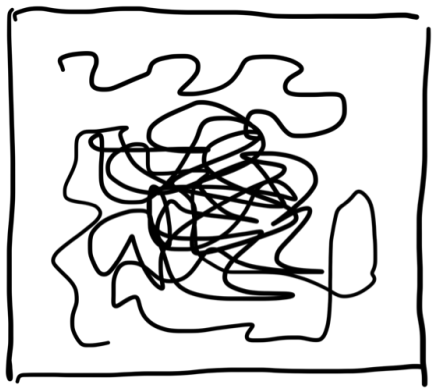
$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} f(X_n) \overset{\text{a.s.}}{=} \int f(x)\, \mu(dx).$$

This gives an algorithm you can implement by truncating the limit at some large N



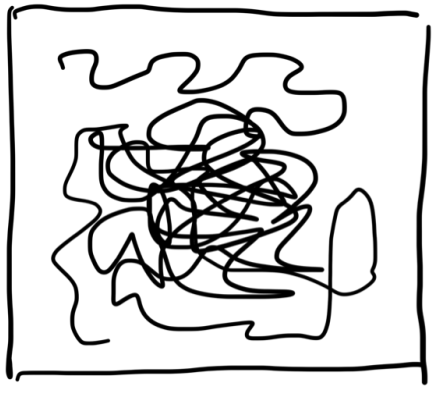
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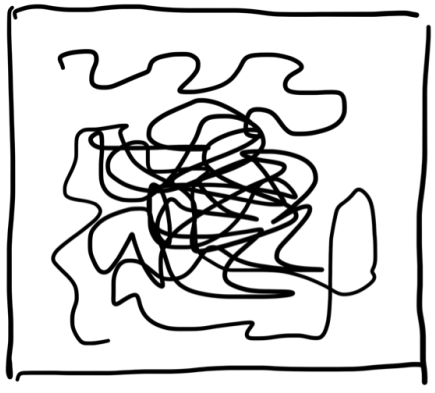


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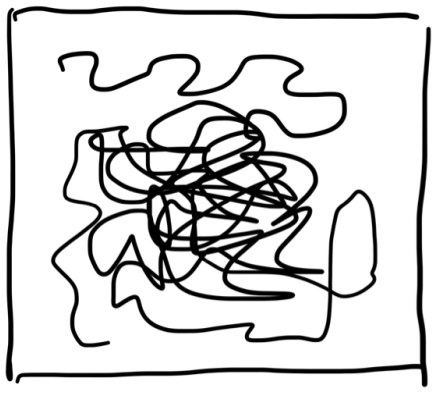
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Proving one of these is the key task when designing an MCMC algorithm

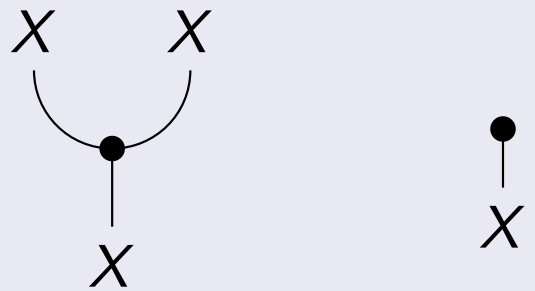
Basic MCMC in Markov categories

Review of Markov categories

These concepts can be considered naturally in terms of Markov categories

Definition ([Fritz, 2020], [Cho and Jacobs, 2019])

A Markov category is a semicartesian symmetric monoidal category $(\mathbf{C}, \otimes, \mathcal{I})$ in which every object $\mathcal{X}$ is equipped with a suitably compatible commutative comonoid structure $(\mathrm{copy}_{\mathcal{X}}, \mathrm{del}_{\mathcal{X}})$, denoted in string diagrams by

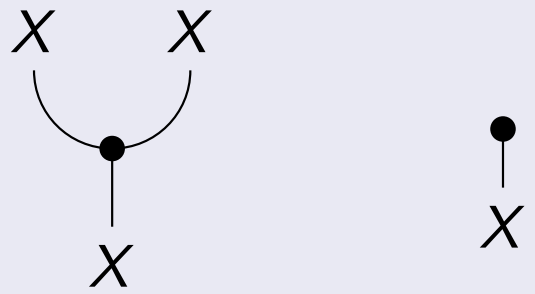


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Basically, Markov categories abstract how information flows in systems possibly involving stochasticity or nondeterminism

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A key example is **Stoch**, the Markov category of measurable spaces and Markov kernels

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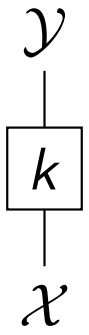
So morphisms $P : \mathcal{X} \to \mathcal{Y}$ are "really" functions $P : \mathcal{X} \times \Sigma_{\mathcal{Y}} \to [0, 1]$

Markov kernels as a Markov category

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So morphisms $P : \mathcal{X} \to \mathcal{Y}$ are "really" functions $P : \mathcal{X} \times \Sigma_{\mathcal{Y}} \to [0, 1]$

Think of P as a "function $\mathcal{X} \to \mathcal{Y}$ with stochastic output"



Composition

Given Markov kernels $P : \mathcal{X} \to \mathcal{Y}$ and $Q : \mathcal{Y} \to \mathcal{Z}$, their sequential composition $Q \circ P : \mathcal{X} \to \mathcal{Z}$ is defined via Chapman-Kolmogorov:

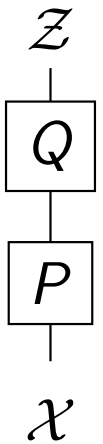
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Depict this as



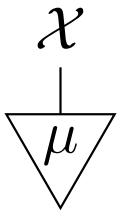
The unit object

In **Stoch**, the unit object $\mathcal{I} \cong \{\bullet\}$ is the singleton measurable space

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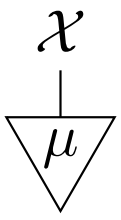
Markov kernels $\mu : \mathcal{I} \to \mathcal{X}$ encode a single probability measure $\mu(\bullet, \mathrm{d}x)$ on $\mathcal{X}$, depicted as:



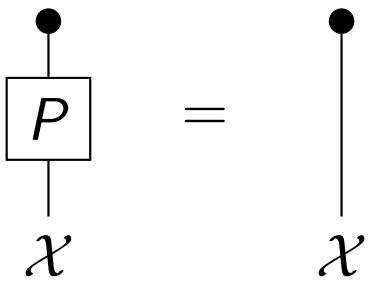
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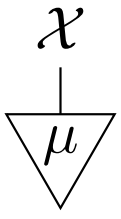
There is exactly one Markov kernel $\mathcal{X} \to \mathcal{I}$, namely $\mathrm{del}_{\mathcal{X}}$. As a result:



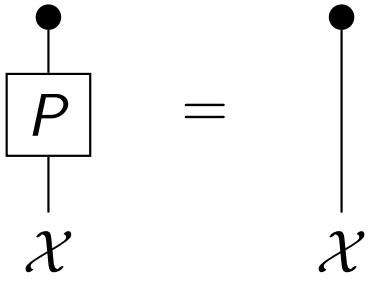
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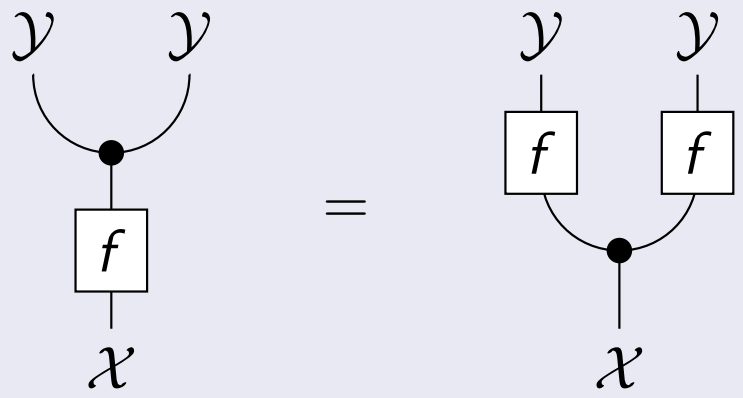
Markov categories basically axiomatise these (and other) behaviours

Determinism

Markov categories model stochastic processes, but as an important special case, we have the following:

Definition (Fritz [2020])

A morphism $f : \mathcal{X} \to \mathcal{Y}$ in a Markov category $\mathbf{C}$ is deterministic if

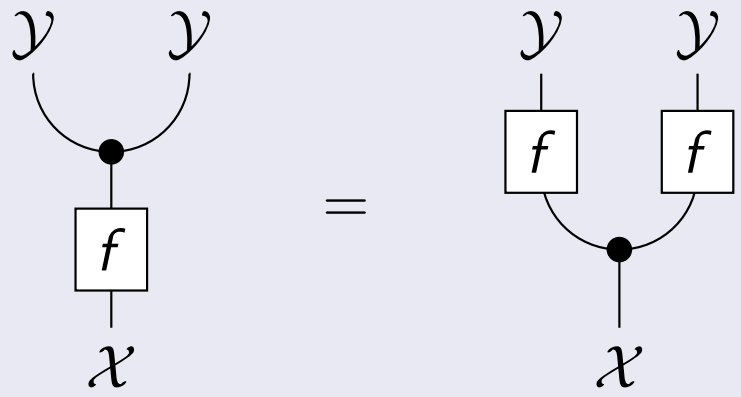


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Example

In **Stoch**, when $\mathcal{X}$ and $\mathcal{Y}$ are standard Borel, these correspond to measurable functions $\mathcal{X} \to \mathcal{Y}$.

Invariance and reversibility

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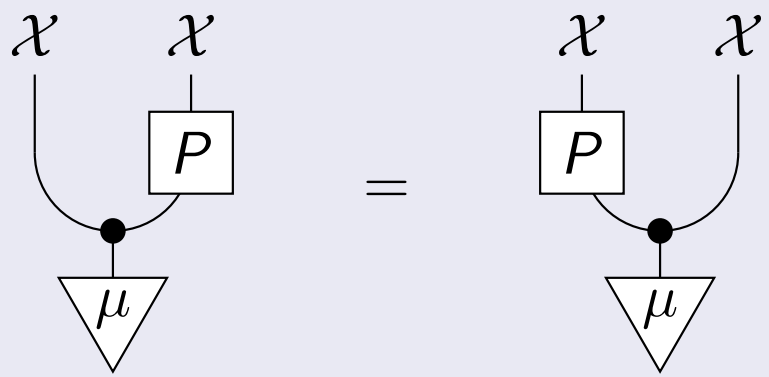
Definition

Let $\mu : \mathcal{I} \to \mathcal{X}$ and $P : \mathcal{X} \to \mathcal{X}$ be morphisms in a Markov category $\mathbf{C}$. We say that P is μ-invariant if

$$P \circ \mu = \mu.$$

Definition

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(1)

Reversibility implies invariance

We can now reason about these concepts without measure theory:

Proposition

If $P : \mathcal{X} \to \mathcal{X}$ is μ-reversible, then P is μ-invariant.

Reversibility implies invariance

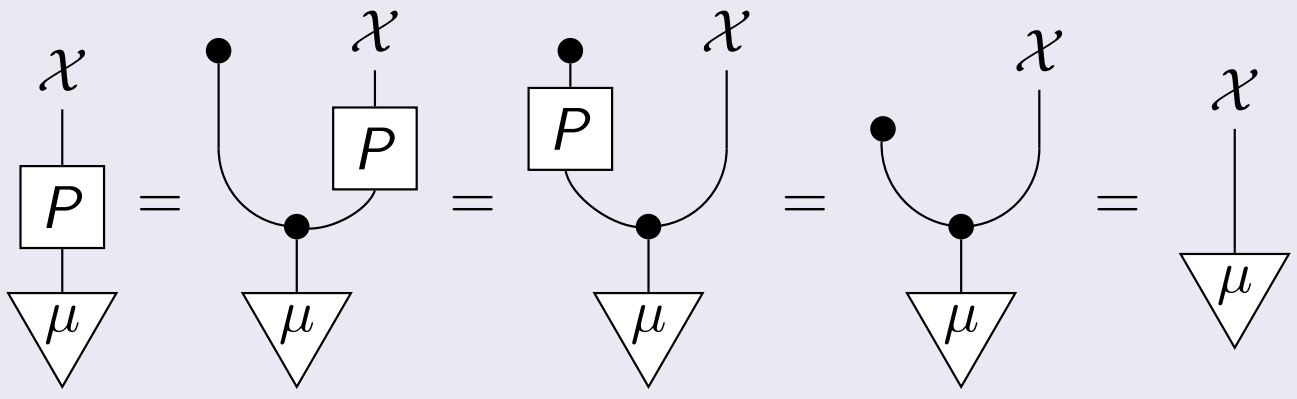
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Proof.

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The converse direction

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However:

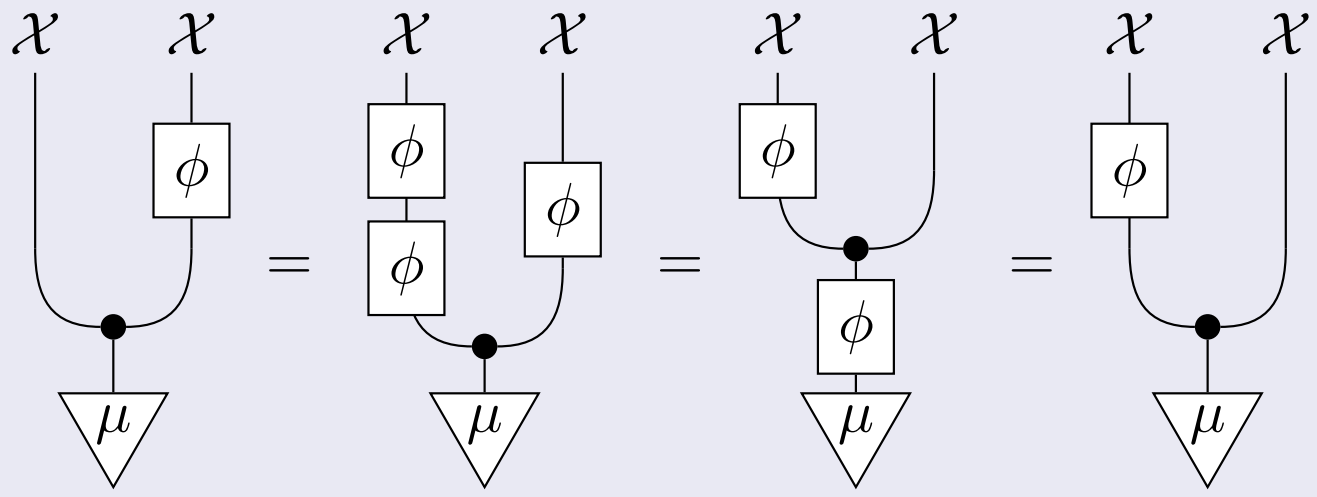
Proposition

Let $\mu : \mathcal{I} \to \mathcal{X}$ and $\phi : \mathcal{X} \to \mathcal{X}$ be morphisms in a Markov category $\mathbf{C}$, where ϕ is a deterministic involution (i.e. $\phi \circ \phi = \mathrm{id}$). If ϕ is μ-invariant, then it is μ-reversible.

Proof

Proof.

If ϕ is μ-invariant, then we have the following:

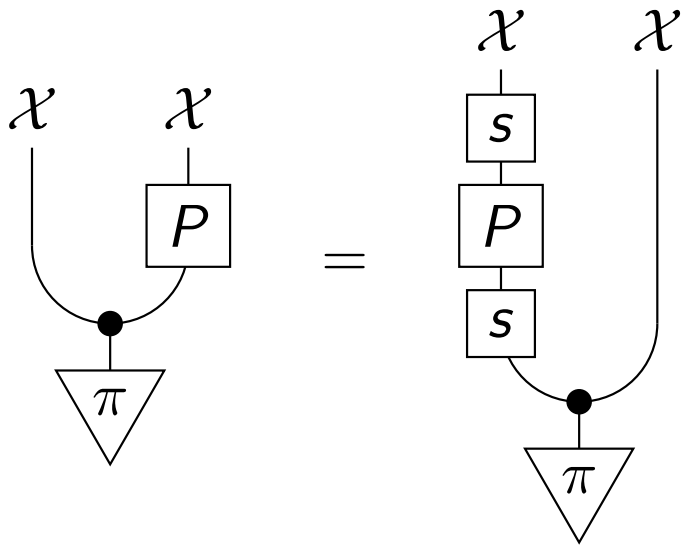


where the first step uses the involution property, the second uses determinism, and the third uses invariance. □

Other things we can talk about

Using Markov categories, we can also reason about:

- Skew reversibility



(2)

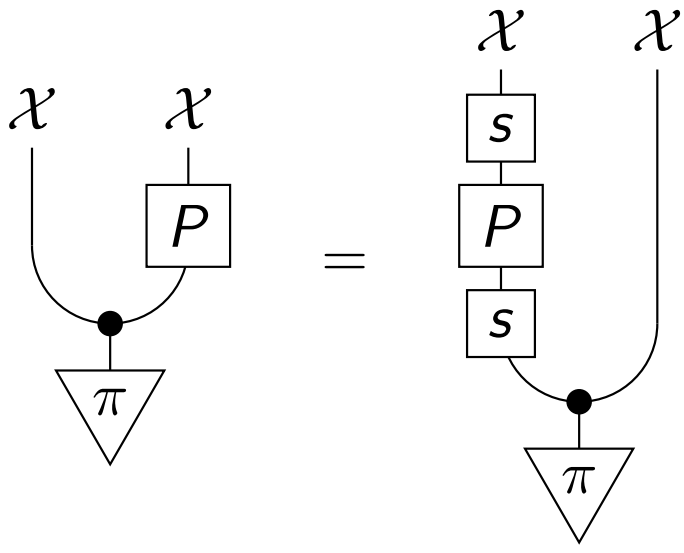
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$$\mathcal{X} \xrightarrow{R} \mathcal{E} \xrightarrow{P} \mathcal{E} \xrightarrow{R^\dagger} \mathcal{X}.$$

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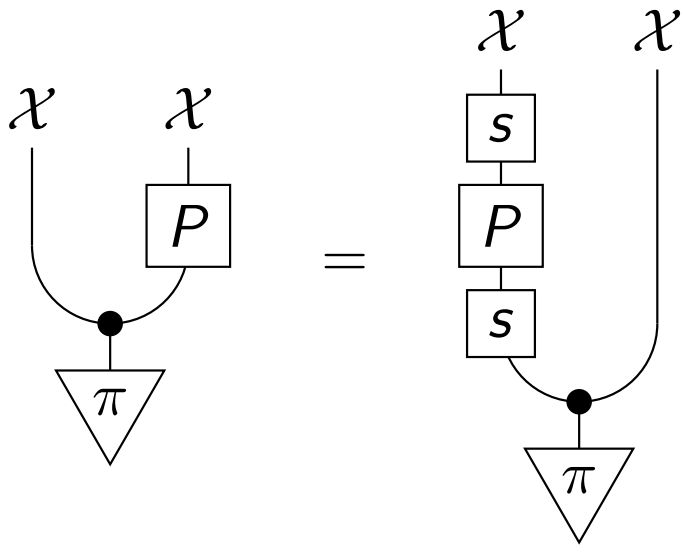
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The Metropolis-Hastings algorithm

The methodological problem of MCMC

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Problem

Given a target distribution μ on $\mathcal{X}$, how do we construct a μ-invariant Markov kernel $P : \mathcal{X} \to \mathcal{X}$?

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Recent efforts has sought to unify these within an involutive framework

Involutive Metropolis-Hastings

A key paper is [Andrieu et al., 2020], who consider Markov kernels of form:

$$\Pi_{\mathrm{MH}}(\xi, \mathrm{d}\xi') := \alpha(\xi)\, \delta_{\phi(\xi)}(\mathrm{d}\xi') + (1 - \alpha(\xi))\, \delta_{\xi}(\mathrm{d}\xi'),$$

where $\phi : \mathcal{E} \to \mathcal{E}$ is a measurable involution, $\alpha : \mathcal{E} \to [0, 1]$ is an acceptance probability, and δ is a Dirac delta

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In practice this is combined with a state space augmentation step

Reversibility result

Theorem (Theorem 3 of Andrieu et al. [2020])

Suppose μ is a finite measure on a measurable space $\mathcal{E}$, and $\phi : \mathcal{E} \to \mathcal{E}$ is a measurable involution. Then there exists a measurable $S \subseteq \mathcal{E}$ such that the restriction $\mu|_S$ and its pushforward $\mu|_S^\phi$ are equivalent as measures, and $\mu|_{S^c}$ and its pushforward $\mu|_{S^c}^\phi$ are singular. The kernel Π_{MH} above is then $\mu|_S$-reversible if

$$\alpha(\xi) = \begin{cases} \alpha(\phi(\xi))\, r(\xi) & \text{if } \xi \in S, \\ 0 & \text{otherwise.} \end{cases}$$

where $r := \frac{\mathrm{d}\mu|_S^\phi}{\mathrm{d}\mu|_S}$ denotes the Radon–Nikodym derivative.

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Proof requires many lines of technical measure-theoretic details

Question

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Usually reversibility is proven for each term separately. However, these are not Markov kernels! (They do not integrate to 1.)

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What structure is needed to prove this result synthetically?

Could try Markov categories, but one issue: the kernel has form

$$\Pi_{\mathrm{MH}}(\xi, \mathrm{d}\xi') = \alpha(\xi)\, \delta_{\phi(\xi)}(\mathrm{d}\xi') + (1 - \alpha(\xi))\, \delta_{\xi}(\mathrm{d}\xi').$$

Usually reversibility is proven for each term separately. However, these are not Markov kernels! (They do not integrate to 1.)

We therefore move to copy-discard (CD) categories, where morphisms need not be normalised.

A review of CD categories

Definition (Cho and Jacobs [2019])

A copy-discard (CD) category is a symmetric monoidal category $(\mathbf{C}, \otimes, \mathcal{I})$ where all objects are equipped commutative comonoid structure $(\mathrm{copy}_{\mathcal{X}}, \mathrm{del}_{\mathcal{X}})$ that is suitably compatible with the monoidal structures.

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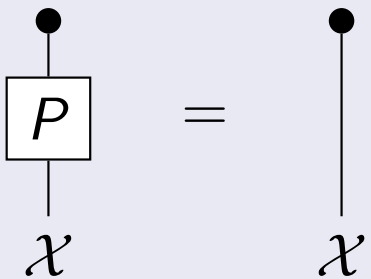
Otherwise very similar (and can import previous definitions)

Normalisation

Practically, dropping semicartesianness amounts to saying that morphisms need not be normalised:

Definition

A morphism $P : \mathcal{X} \to \mathcal{Y}$ in a CD category $\mathbf{C}$ is normalised if

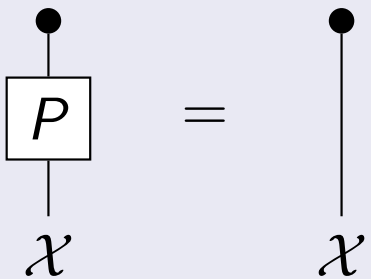


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In a Markov category, every morphism is normalised. In a CD category, these form a Markov category

$$\mathbf{C}_{\mathrm{nrm}} \subseteq \mathbf{C}$$

s-finite kernels

A key example of a CD category is **sfKern**, the category of measurable spaces and s-finite kernels

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An s-finite kernel $K : \mathcal{X} \to \mathcal{Y}$ is like a Markov kernel, but allows

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(Formally, $K = \sum_{n=1}^{\infty} K_n$ where each $\sup_x K_n(x, \mathcal{Y}) < \infty$.)

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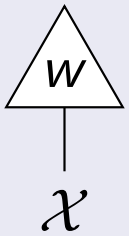
(Formally, $K = \sum_{n=1}^{\infty} K_n$ where each $\sup_x K_n(x, \mathcal{Y}) < \infty$.)

We recover $\mathbf{Stoch} = \mathbf{sfKern}_{\mathrm{nrm}}$

Effects

Definition

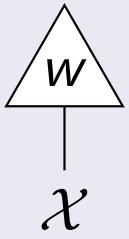
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Example

In **sfKern**, an effect $w : \mathcal{X} \to \mathcal{I}$ corresponds to the function $w : \mathcal{X} \to [0, \infty]$ defined as $w(x) \coloneqq w(x, \mathcal{I})$. Also,

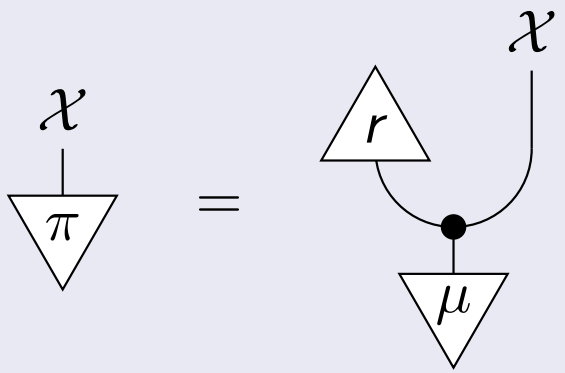
$$(w \circ K)(x) = \int w(y)\, K(x, \mathrm{d}y),$$

i.e. composing with w corresponds to integration against this function.

Radon-Nikodym derivatives (i.e. densities)

Definition

Let $\pi, \mu : \mathcal{I} \to \mathcal{X}$ be morphisms in a CD category $\mathbf{C}$. A Radon–Nikodym derivative of π with respect to μ is an effect $r \equiv \frac{\mathrm{d}\pi}{\mathrm{d}\mu}$ such that

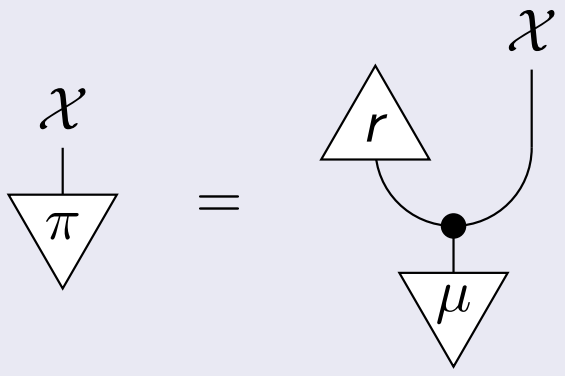


(3)

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(3)

Example

In **sfKern**, the condition (3) holds if and only if we always have

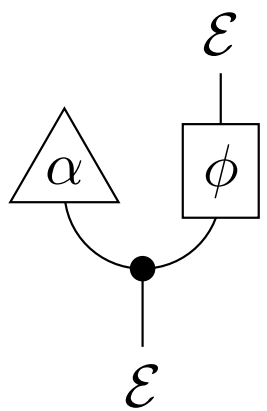
$$\pi(A) = \int_A r(x)\, \mu(\mathrm{d}x)$$

The first MH summand

Recall that MH kernel is

$$\Pi_{\mathrm{MH}}(\xi, \mathrm{d}\xi') = \alpha(\xi)\, \delta_{\phi(\xi)}(\mathrm{d}\xi') + (1 - \alpha(\xi))\, \delta_{\xi}(\mathrm{d}\xi').$$

This is a morphism in **sfKern**. In string diagrams, the red term becomes

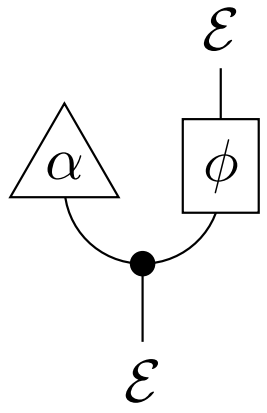


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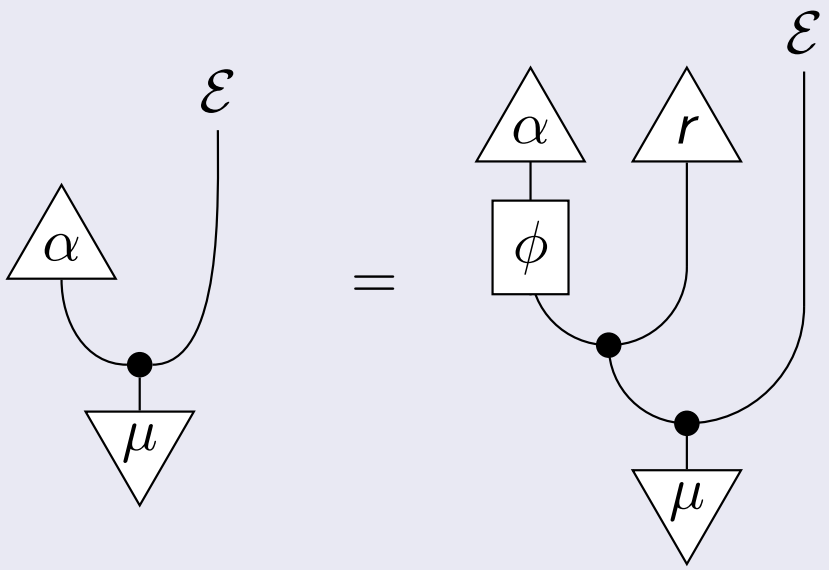


So we can ask: when is this μ-reversible?

Reversibility of the first MH summand

Theorem

Suppose $\mu : \mathcal{I} \to \mathcal{E}$, $\phi : \mathcal{E} \to \mathcal{E}$, *and* $\alpha : \mathcal{E} \to \mathcal{I}$ *are morphisms in a CD category* $\mathbf{C}$, *where* ϕ *is a deterministic involution, and there exists* $r = \frac{\mathrm{d}(\phi \circ \mu)}{\mathrm{d}\mu}$. *Then the morphism above is* μ*-reversible if and only if*

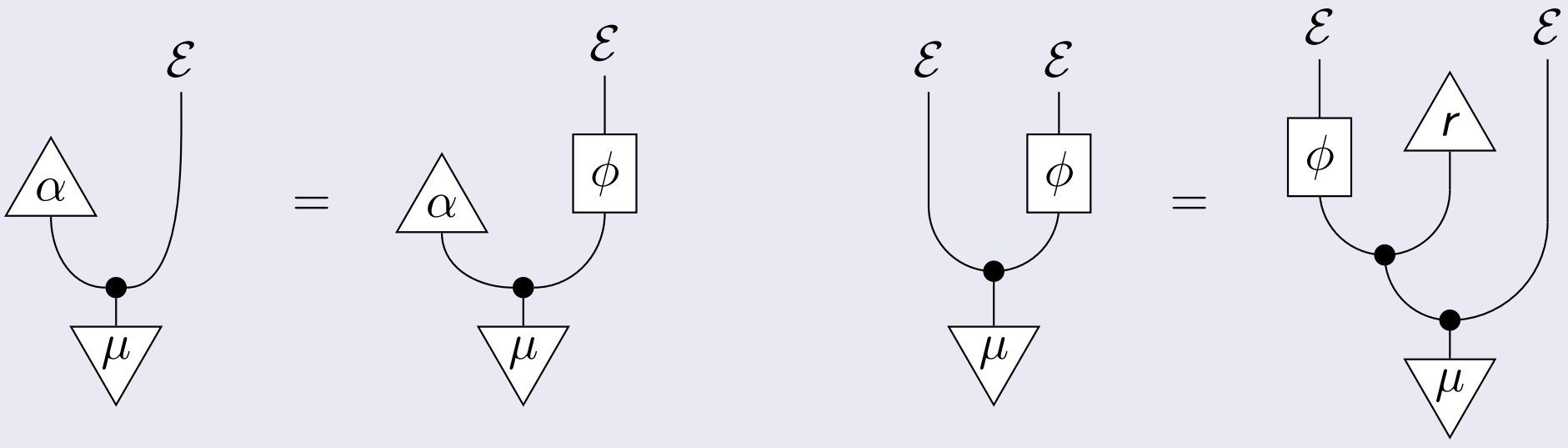


(4)

Proof

Proof.

By string diagram manipulations, reversibility is equivalent to first equation below. Extending our earlier argument about reversibility of involutions gives the second equation (always):



Now apply $\alpha : \mathcal{E} \to \mathcal{I}$ to both left-hand wires. □

Next steps

What about the full kernel?

$$\Pi_{\mathrm{MH}}(\xi, \mathrm{d}\xi') = \alpha(\xi)\, \delta_{\phi(\xi)}(\mathrm{d}\xi') + (1 - \alpha(\xi))\, \delta_{\xi}(\mathrm{d}\xi').$$

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2. $\alpha(\xi)$ and $1 - \alpha(\xi)$

For full result of Andrieu et al. [2020], also need to decompose μ into two parts, $\mu|_S$ and $\mu|_{S^c}$, and talk absolute continuity and singularity of these

Semiadditive CD categories

Key idea

A natural place to start is to enrich over commutative monoids

Definition

A CD category is semiadditive if each hom-set $\mathbf{C}(\mathcal{X}, \mathcal{Y})$ is equipped with a commutative monoid structure $(+, 0)$ such that

$$\begin{aligned} P \circ 0 &= 0 & P \circ (Q + R) &= P \circ Q + P \circ R \\ 0 \circ P &= 0 & (Q + R) \circ P &= Q \circ P + R \circ P. \end{aligned} \tag{5}$$

and also

$$\begin{aligned} P \otimes 0 &= 0 & P \otimes (Q + R) &= (P \otimes Q) + (P \otimes R) \\ 0 \otimes P &= 0 & (Q + R) \otimes P &= (Q \otimes P) + (R \otimes P). \end{aligned} \tag{6}$$

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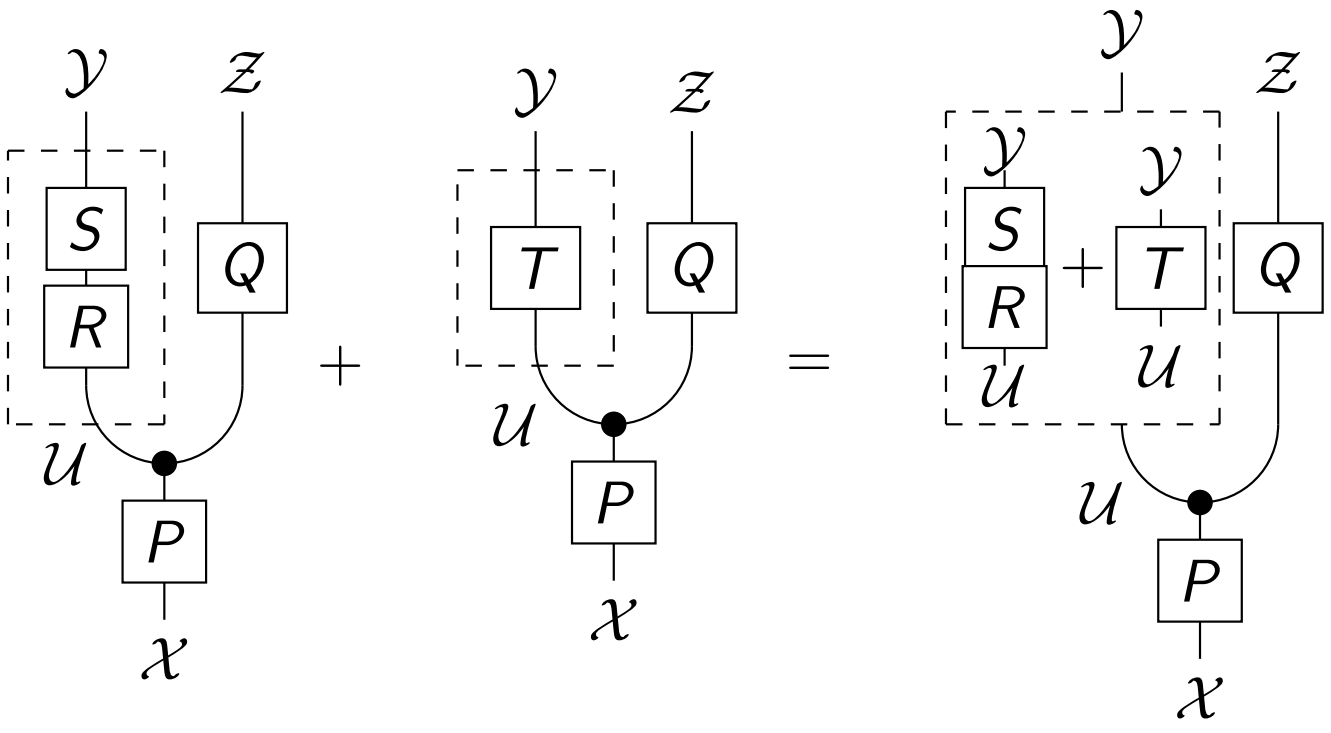
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Note: in isolation, various parts below do not use the full CD structure (just the **CMon**-enrichment)

Graphical interpretation

Graphically, semiadditivity means we can sum morphisms by "pattern matching" on their string diagrams:



Also, if any box is 0, then the whole diagram is 0.

Key example

Example

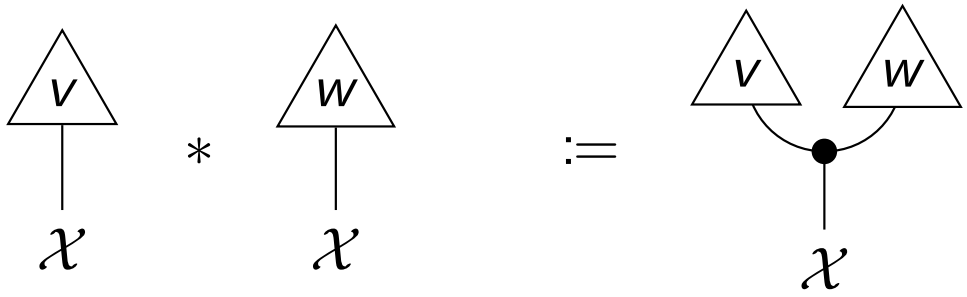
Let $\mathcal{X}$ and $\mathcal{Y}$ be measurable spaces, and define the zero kernel $0 : \mathcal{X} \to \mathcal{Y}$ and the sum $P + Q : \mathcal{X} \to \mathcal{Y}$ of kernels $P, Q : \mathcal{X} \to \mathcal{Y}$ via:

$$0(x, \mathrm{d}y) \coloneqq 0$$
$$(P + Q)(x, \mathrm{d}y) \coloneqq P(x, \mathrm{d}y) + Q(x, \mathrm{d}y).$$

This equips **sfKern** with the structure of a semiadditive CD category.

The semiring of effects

For any CD category $\mathbf{C}$, the effects $\mathbf{C}(\mathcal{X}, \mathcal{I})$ admit another commutative monoid structure $(*, 1)$ [Fritz et al., 2023a]: multiplication is



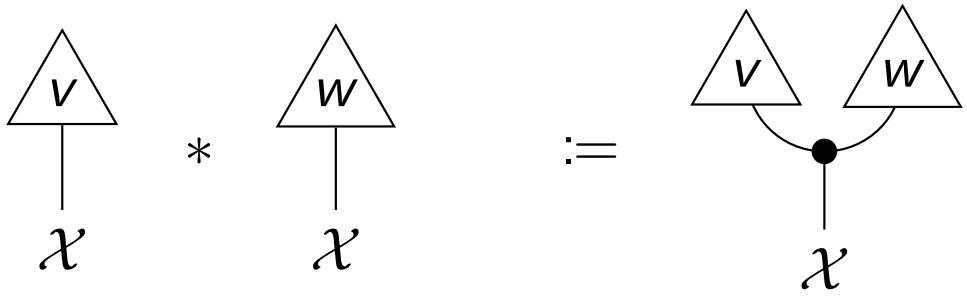
(7)

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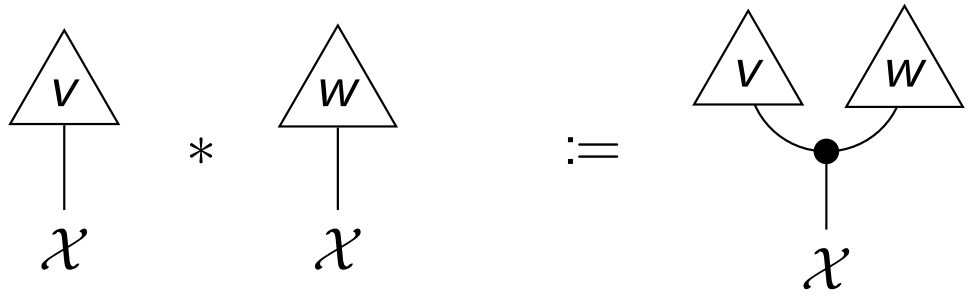
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When $\mathbf{C}$ is semiadditive, this is in fact a commutative semiring

Enrichment over preorders

Recall that every commutative monoid M admits a canonical preorder $\leq$ defined by

$$a \leq b \iff a + c = b \text{ for some } c \in M.$$

This extends to morphisms in a semiadditive CD category

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Also easy to check that this is in fact a monoidal enrichment over preorders

Compatibility conditions

Can introduce various compatibility conditions, e.g.:

- $\mathbf{C}$ is *zero-sum-free* if $P + Q = 0$ implies $P = Q = 0$
- $\mathbf{C}$ is *zero-monic* [Cho et al., 2015] if $1 \circ P = 0$ implies $P = 0$.
- $\mathbf{C}$ has *no zero divisors* if $P \otimes Q = 0$ implies $P = 0$ or $Q = 0$

Probabilistic concepts in semiadditive CD categories

Substochasticity

Definition

A morphism $P : \mathcal{X} \to \mathcal{Y}$ in a semiadditive CD category $\mathbf{C}$ is substochastic if

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Proposition

Let $\mathbf{C}$ be a semiadditive CD category. Then the objects and substochastic morphisms of $\mathbf{C}$ form a CD category.

Convex combinations

Given a probability $\alpha : \mathcal{X} \to \mathcal{I}$, have another probability α^c such that

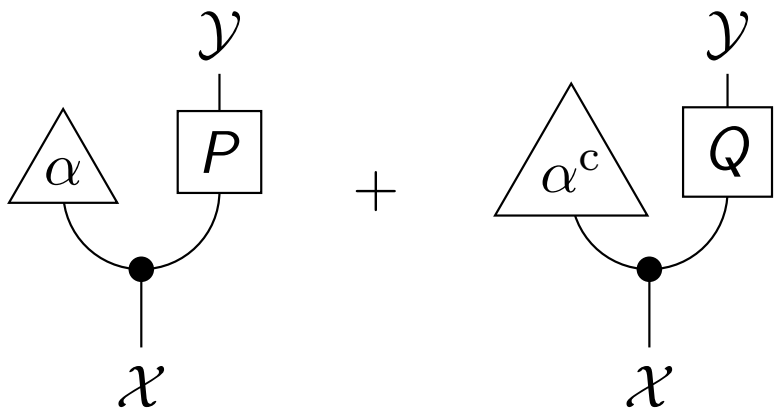
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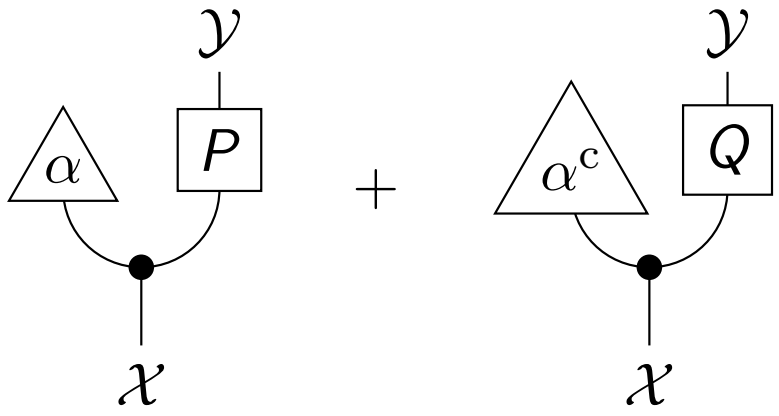
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(8)

Example

In **sfKern**, this is the kernel $\alpha(x)\, P(x, dy) + (1 - \alpha(x))\, Q(x, dy)$.

Aside: convex combinations vs. random choice

Consider again the convex combination

$$\alpha(x)\, P(x, dy) + (1 - \alpha(x))\, Q(x, dy) \tag{9}$$

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Statisticians are trained to interpret this as a random choice process like

$$B \sim \mathrm{flip}(\alpha(x)) \qquad Y \sim P(x, \mathrm{d}y) \qquad Y' \sim Q(x, \mathrm{d}y)$$
$$\text{return } Y \text{ if } B = 1 \text{ else } Y' \tag{10}$$

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What morphism is flip? What is the if/else statement?

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So in general, (9) and (10) are distinct concepts.

Aside: connection with distributive Markov categories

Recently Ackerman et al. [2024] introduced distributive Markov categories, which do support reasoning about random choice

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Proposition

Let $\mathbf{C}$ be a semiadditive CD category with an initial object and binary coproducts whose inclusions are normalised and deterministic. These equip $\mathbf{C}_{\mathrm{nrm}}$ with the structure of a distributive Markov category.

Cancellativity

Definition

Let $P : \mathcal{X} \to \mathcal{Y}$ be a morphism in a semiadditive CD category $\mathbf{C}$. We say that P is *cancellative* if

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What does this mean in **sfKern**? Recall that a measure μ is σ-finite if we can write

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Example

Let $K : \mathcal{X} \to \mathcal{Y}$ be a kernel. If every $K(x, \mathrm{d}y)$ is σ-finite, then K is cancellative. The converse holds e.g. if singletons in $\mathcal{X}$ are measurable.

Finiteness

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Definition

A semiadditive CD category $\mathbf{C}$ is finitely cancellative if all finite morphisms in $\mathbf{C}$ are cancellative.

The full Metropolis-Hastings kernel

Recap

Recall that the full involutive Metropolis-Hastings kernel is

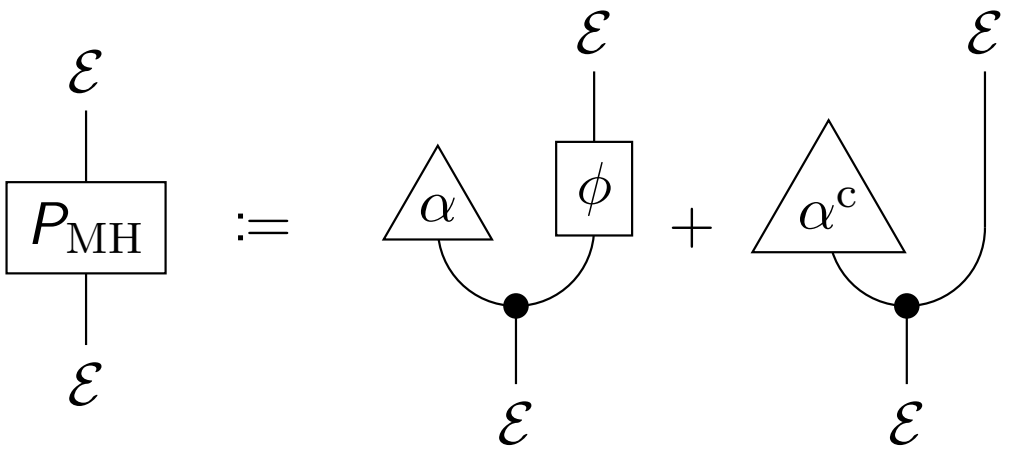
$$\Pi_{\mathrm{MH}}(\xi, \mathrm{d}\xi') = \alpha(\xi)\, \delta_{\phi(\xi)}(\mathrm{d}\xi') + (1 - \alpha(\xi))\, \delta_{\xi}(\mathrm{d}\xi').$$

Recap

Recall that the full involutive Metropolis-Hastings kernel is

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In a semiadditive CD category $\mathbf{C}$, we can consider instead



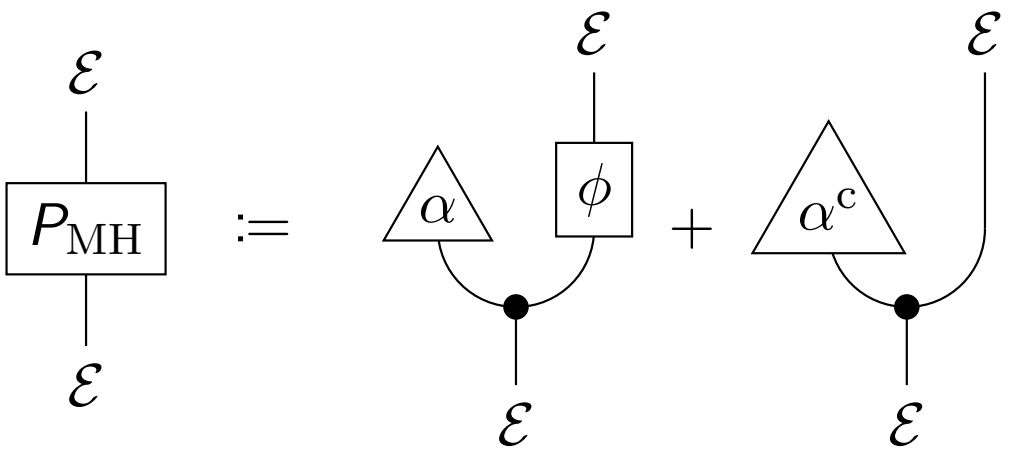
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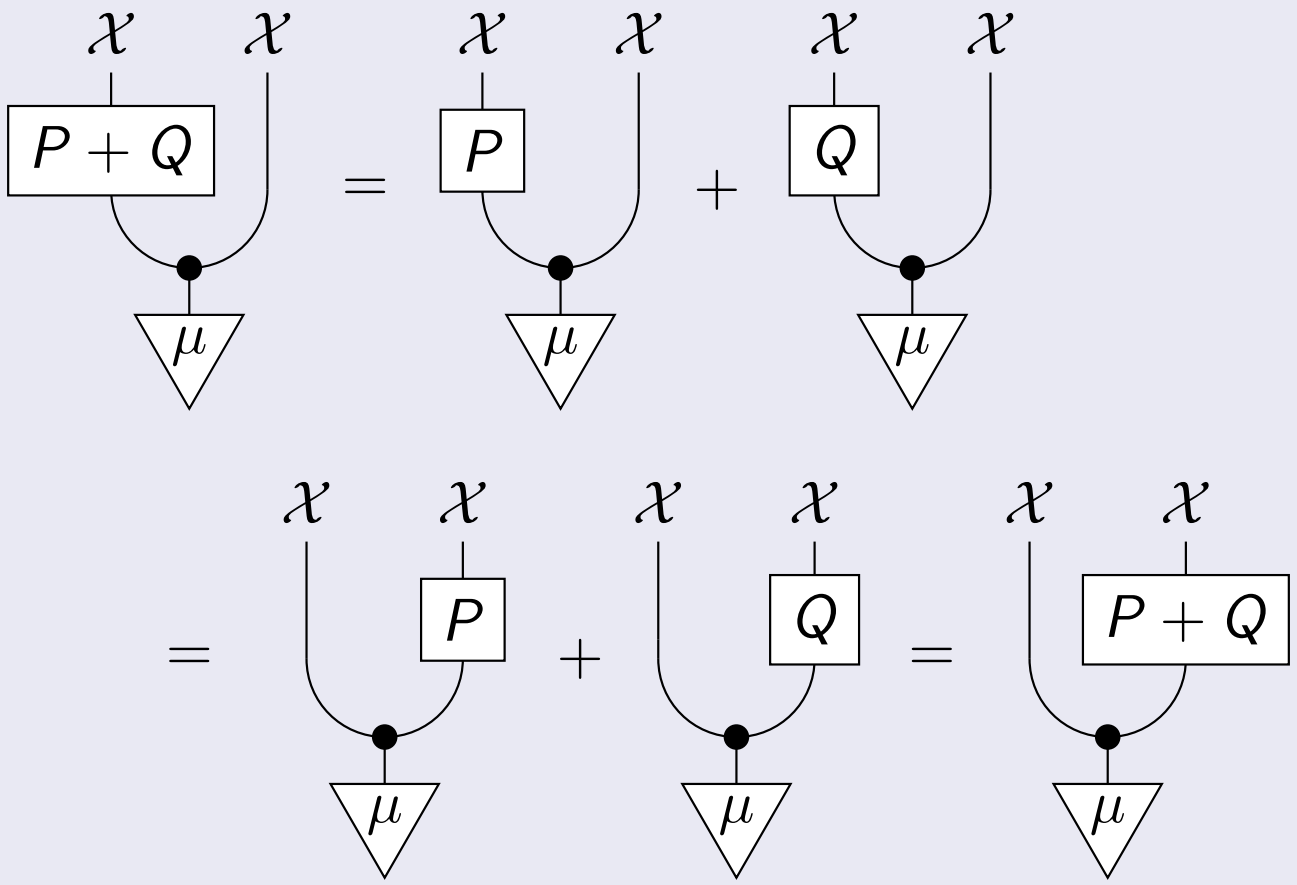
We already gave necessary and sufficient conditions for the first summand to be μ-reversible (and second summand is trivially reversible)

Reversibility of sums (1)

Proposition

If P and Q and are μ-reversible, then so is $P + Q$.

Proof.



Reversibility of sums (2)

For the converse direction, we also have the following:

Proposition

Suppose $\mathbf{C}$ is finitely cancellative, μ is finite, and Q is substochastic. If $P + Q$ and Q are both μ-reversible, then so is P.

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Proof is similar (but now we cancel the term involving Q on both sides)

Application to Metropolis-Hastings

Theorem

Let $\mathbf{C}$ *be a finitely cancellative semiadditive CD category, and suppose*

- $\mu : \mathcal{I} \to \mathcal{E}$ *is finite*
- $\phi : \mathcal{E} \to \mathcal{E}$ *is a deterministic involution*
- $\alpha : \mathcal{E} \to \mathcal{I}$ *is a probability*
- $r = \frac{\mathrm{d}(\phi \circ \mu)}{\mathrm{d}\mu}$ *is a Radon–Nikodym derivative.*

Then P_{MH} *is* μ*-reversible if and only if the following "balancing condition" holds:*

$$\alpha = (\alpha \circ \phi) * r \qquad \mu\text{-almost everywhere.} \tag{11}$$

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Proof.

We already gave necessary and sufficient condition for the first term, which is exactly (11). The result now follows by the previous two results. □

Theorem

Instantiating this in **sfKern** gives the following:

Corollary

Suppose μ is a finite measure on a measurable space $\mathcal{E}$, and $\phi : \mathcal{E} \to \mathcal{E}$ is a measurable involution such the pushforward measure μ^ϕ is absolutely continuous with respect to μ. Then the MH kernel Π_{MH} is μ-reversible if and only if

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Also can extend immediately to skew-reversibility case

Absolute continuity and Lebesgue decompositions

Motivation

The theorem of Andrieu et al. [2020] also involved a decomposition step:

Theorem

Suppose μ is a finite measure on a measurable space $\mathcal{E}$, and $\phi : \mathcal{E} \to \mathcal{E}$ is a measurable involution. Then there exists a measurable $S \subseteq \mathcal{E}$ such that the restriction $\mu|_S$ and its pushforward $\mu|_S^\phi$ are equivalent as measures, and $\mu|_{S^c}$ and its pushforward $\mu|_{S^c}^\phi$ are singular.

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How can we reason about this in a semiadditive CD category?

Recap of absolute continuity

Given measures μ and ν on a space $\mathcal{X}$, recall that ν is absolutely continuous with respect to μ if

$$\mu(A) = 0 \implies \nu(A) = 0.$$

Often written as $\nu \ll \mu$.

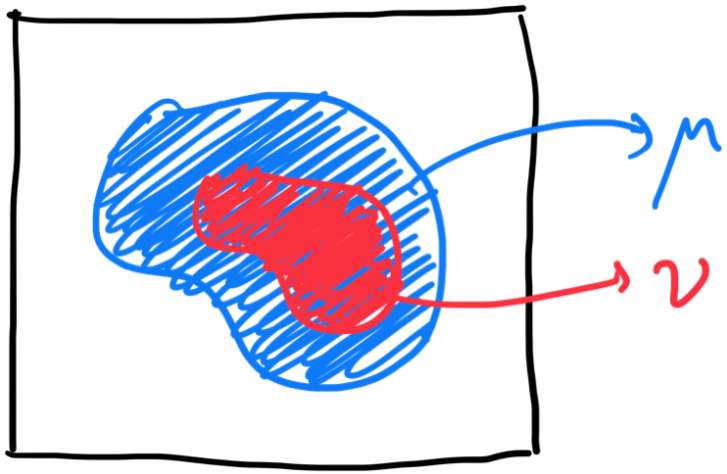
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Conceptually this looks as follows:



Also, ν and μ are equivalent if both are absolutely continuous with respect

Another preorder

Definition

Let $P, Q : \mathcal{X} \to \mathcal{Y}$ be morphisms in a semiadditive CD category $\mathbf{C}$. We write $P \ll Q$ if

$$S \circ Q \circ R = 0 \implies S \circ P \circ R = 0. \tag{12}$$

We write $P \equiv Q$ if both $P \ll Q$ and $Q \ll P$ hold.

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In **sfKern**, we have $P \ll Q$ iff $P(x, \mathrm{d}y)$ is absolutely continuous with respect to $Q(x, \mathrm{d}y)$ for all $x \in \mathcal{X}$.

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Differs from the notion of Fritz et al. [2023b], but related when $\mathbf{C}$ is zero-monic and has no zero divisors

Lebesgue decompositions

A classical measure theory concept is a Lebesgue decomposition: given two measures μ and ν, this is a decomposition

$$\nu = \nu_{\mathrm{ac}} + \nu_{\mathrm{s}}$$

where here:

1. $\nu_{\mathrm{ac}} \ll \mu$
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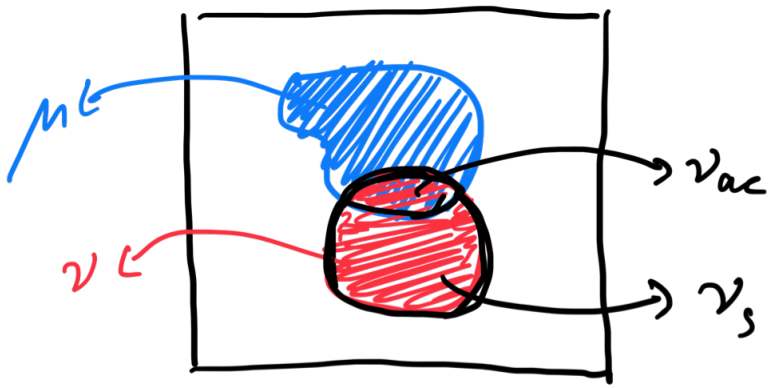
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Note that meets do not have to exist, and when they do are not usually unique

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Suppose $P, Q : \mathcal{X} \to \mathcal{Y}$ are kernels. If $P(x, \mathrm{d}y)$ and $Q(x, \mathrm{d}y)$ are singular for all $x \in \mathcal{X}$, then $P \perp Q$. The converse also holds when e.g. all singleton subsets of $\mathcal{X}$ are measurable, and P and Q are pointwise σ-finite.

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Let $P, Q : \mathcal{X} \to \mathcal{Y}$ be morphisms in a semiadditive CD category $\mathbf{C}$. A *Q-decomposition* of P is a pair of morphisms $P_{\mathrm{ac}}, P_{\mathrm{s}} : \mathcal{X} \to \mathcal{Y}$ such that

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is a Lebesgue decomposition with respect to $Q(x, \mathrm{d}y)$ for all $x \in \mathcal{X}$, then $(P_{\mathrm{ac}}, P_{\mathrm{s}})$ is a Q-decomposition of P. The converse holds when all singleton subsets of $\mathcal{X}$ are measurable, and P_{ac} and P_{s} are pointwise σ-finite.

Completing the result of Andrieu et al. [2020]

Theorem (Proof omitted)

Let $\mathbf{C}$ *be a zero-sum-free semiadditive CD category. Let* $P : \mathcal{X} \to \mathcal{Y}$ *and* $\phi : \mathcal{Y} \to \mathcal{Y}$ *be morphisms in* $\mathbf{C}$*, where* ϕ *is an involution, and suppose* $(P_{\mathrm{ac}}, P_{\mathrm{s}})$ *is a* $(\phi \circ P)$*-decomposition of* P*. Then we have both the following:*

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Corollary (uses existence of Lebesgue decompositions in **sfKern**)

Suppose μ is a σ-finite measure on $\mathcal{E}$, and $\phi : \mathcal{E} \to \mathcal{E}$ is a measurable involution. Then there exists a measurable $S \subseteq \mathcal{E}$ such that the restriction $\mu|_S$ and its pushforward $\mu|_S^\phi$ are equivalent, and $\mu|_{S^c}$ and its pushforward $\mu|_{S^c}^\phi$ are singular.

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This allows us to say many things without measure theory!

Thank you!

Advertisement

Please do get in touch if you're interested in working together on anything related to categorical probability, statistics, or machine learning!

I will also likely be hiring postdocs and have PhD scholarships available in the next 6-12 months, so please reach out if you're interested in those as well!

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE



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Appendix

Metropolis-Hastings

Key requirements:

- Target density $\mu(x)$
- Proposal density $q(y|x)$ (gives a Markov kernel $Q : \mathcal{X} \to \mathcal{X}$)

Algorithm

Given input $x \in \mathcal{X}$, do the following:

1. Sample $Y \sim q(\cdot|x)$
2. With probability $\alpha(x, Y) = \min\left(1, \frac{\mu(Y)q(x|Y)}{\mu(x)q(Y|x)}\right)$, return Y
3. Otherwise, return x

This gives a kernel written as

$$P(dy|x) = \alpha(x, y)\, Q(dy|x) + \left(1 - \int \alpha(x, y)\, Q(dy|x)\right) \delta_x(dy)$$

where δ is the Dirac delta